Automatic Segmentation of Time Series Data with PELT Algorithm for Predictive Maintenance in the Flat Steel Industry

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***Abstract*— In this study, we aim to test the usability of Change Point Detection (CPD) algorithms (specifically the Pruned Exact Linear Time-PELT) to facilitate the utilization of large volumes of data within predictive mechanisms in the industry. We proposed an efficient CPD parameter selection mechanism for defect diagnosis using time-series vibration data from critical assets. We emphasized the practical algorithm PELT to ensure broad industrial applicability. Our experimental analysis, using synthetic and actual vibration data, demonstrated the practical applicability and effectiveness of PELT algorithm for automatic segmentation. The numerical results show the potential of CPD methodologies for improving predictive maintenance operations by providing an automatic segmentation mechanism. This pipeline proposes a way to increase the operational efficiency and scalability of predictive maintenance approaches, enhancing maintenance procedures and ensuring the long-term reliability of industrial systems.**

Keywords*— *Predictive Maintenance, Fault Detection, Change Point Detection (CPD), Real-Time Monitoring, Vibration Data, Cost-Effective Solutions, Hybrid Monitoring System, Scalability, Operational Efficiency

I. Introduction

Predictive maintenance, focusing on early and accurate fault detection, has become increasingly vital in the manufacturing industry. The field of predictive maintenance could be categorized by the following levels of maturity within the context of manufacturing industry: **a)** Level 1 involves visual inspections, where experts perform periodic physical inspections and conclusions are based on the expertise of the maintenance personnel. **b)** Level 2 consists of instrument inspections, where conclusions are drawn from a combination of periodic physical expert inspections and instrument read-outs. **c)** Level 3 is real-time condition monitoring, where continuous monitoring is conducted, and alerts are given based on pre-established rules. **d)** Level 4 is predictive maintenance, where historical and current data are stored to predict machine failures using machine learning algorithms. One of the main challenges of predictive maintenance is the quality and quantity of data, which are crucial for analyzing root causes and predicting failures well ahead of time.

Therefore, increasing data quality and coverage is essential for improving the prediction accuracy. Besides, the sensor technology has become more reachable in recent years due to the decrease in the cost of sensor materials. This accessibility has led to widespread adoption of sensors across various industries, enhancing the capability to collect high-quality and continuous data [1]. As a result, there is a growing necessity for a robust infrastructure to ensure this data is structured, continuous, and reliable. On the other hand, most of the collected data remain underutilized due to the lack of corresponding labels. High-quality, labeled data is crucial for training effective machine learning models. This work emphasizes the use of CPD algorithms (mainly the PELT algorithm) to address this problem by automating the labeling process. The high precision of CPD can allow for accurate detection of even minor but significant variations, enabling rapid identification of problems. It significantly enhances the efficiency of predictive maintenance techniques by ensuring early detection of failures and automating the data labeling process, thus guaranteeing uninterrupted industrial operations.

Traditional approaches, while efficient, often lack the precision of CPD in identifying minor modifications that occur before significant problems. CPD enhances predictive maintenance by automatically detecting abrupt changes or deviations in vibration signals, indicating potential issues. By identifying these changes, CPD can effectively help the users to label the massive datasets, making them more useful for predictive maintenance. This automation transforms vast, unlabeled datasets into valuable resources, significantly improving the accuracy of predictions.

This study specifically focused on the flat steel sector, which relies heavily on robust machinery and continuous production. The study was conducted using data obtained from the Borçelik factory (https://www.borcelik.com/en), which specializes in flat steel production. The integration of CPD with traditional fault detection techniques offers a robust, scalable, and cost-effective solution for the manufacturing industry, enhancing operational reliability and productivity.

The rest of this paper is organized as follows; first we included a review of existing literature to represent the current state of CPD methodologies and their applications in predictive maintenance. In the next section, we discussed on the proposed methodology with PELT algorithm on synthetic data that simulate real-world scenarios. This allowed us to fine-tune our models and understand their performance under controlled conditions. Following successful validation on synthetic data, we applied the models to actual vibration data collected from machinery in the flat steel sector. This step was

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crucial in assessing the practical applicability of CPD approaches. Finally, we analyzed the outcomes of applying CPD to real machine data, interpreting the results to understand their implications for predictive maintenance.

II. Literature Review

Predictive maintenance has become an important research and application area in the industrial sector because it can prevent unexpected equipment failures and improve maintenance schedule [2]. A major challenge in the implementation of predictive maintenance is the necessity for large amounts of labeled data, which is essential for supervised learning methodologies. Supervised learning mainly depends on labeled past data to train prediction models [3]. Labeling process is both costly and time-consuming due to the necessity of having subject experts annotate vast quantities of data. In the flat steel business, where equipment runs continually and under various conditions, the task of reliably labeling data may be particularly difficult and costly [4]. Although there have been major advances in machine learning and predictive maintenance, CPD methods have not been generally applied in the current domain [5]. CPD is a method that aims to detect substantial changes in the statistical characteristics of a time-series dataset, which might be suggestive of the beginning of equipment malfunctions. Nevertheless, most standard predictive maintenance models depend on ongoing monitoring and predetermined thresholds to identify abnormalities. Although these approaches can be efficient, they may not have the precision of CPD in detecting minor modifications that occur before significant problems [6]. Another notable challenge is the efficient handling and analysis of the substantial amounts of data produced by industrial machinery [7]. Modern industrial processes have been fitted with many kinds of sensors that consistently gather data, leading to huge quantities of time-series data. Effectively analyzing this data is crucial for precise defect diagnosis and predictive maintenance. Nevertheless, the huge quantity of data might overpower traditional data processing and analysis techniques. To effectively store, evaluate, and derive valuable insights from this data, sophisticated algorithms and significant computational resources are necessary [8]. CPD can offer timely alerts by detecting little yet meaningful variations in sensor measurements, enabling maintenance crews to act before a malfunction develops into a substantial problem. In addition, CPD approaches can be economically advantageous by decreasing reliance on massive, labeled datasets. This is because they can identify changes in the data by analyzing its underlying statistical qualities, rather than depending simply on supervised learning models. CPD techniques are crucial for identifying moments in time where the statistical properties of a time-series data set change significantly [9]. These changes can signal the onset of faults or other significant events in industrial machinery. CPD methods are advantageous because they can detect subtle changes that may not be evident through traditional monitoring techniques [10]. The Cumulative Sum (CUSUM) method is one of the earliest techniques, designed to detect shifts in the mean level of a process by accumulating the sum of deviations from a target value. It flags a change when this sum exceeds a predefined threshold [11]. CUSUM is effective for detecting small shifts in the mean but may not perform well with non-linear or complex changes. Another significant approach is Bayesian approach is Bayesian Change Point Detection, which uses Bayesian inference to estimate the probability of a change point at each time step, providing a probabilistic framework for identifying changes. This method offers flexibility and robustness, accommodating various types of changes and incorporating prior knowledge [12]. However, it can be computationally intensive, especially with large datasets. Kernel CPD applies kernel methods to detect changes in the distribution of data. This technique is particularly effective for identifying non-linear changes that might not be captured by other methods [13]. The selection of appropriate kernel functions and parameters is critical for the performance of this method.

Among the various CPD techniques, the Pruned Exact Linear Time (PELT) algorithm stands out due to its efficiency and accuracy in detecting multiple change points. The PELT algorithm prunes the search space of possible change points, making it computationally efficient while maintaining high accuracy [14]. Therefore, in this work, we focus on the advantages of combining PELT algorithm with traditional fault detection techniques to improve predictive maintenance. While CPD techniques offer significant advantages, they also face several challenges. The effectiveness of CPD methods depends on the quality and granularity of the data; poor data quality can lead to inaccurate detections [15]. Some CPD methods, particularly those involving Bayesian inference or kernel functions, can be computationally demanding, necessitating efficient algorithms and powerful computing resources.

Comprehensive reviews and methodologies existing in the literature, especially focusing on using vibration data and machine learning to predict asset health can be found in [16]. In particular, the study in [17], provides a systematic literature review of machine learning methods applied to predictive maintenance. It explores recent advances and trends in predictive maintenance from a data-driven machine prognostics perspective. The use of supervised learning for predictive maintenance requires a large amount of labeled data, which is expensive and time-consuming to acquire. Additionally, the potential advantages of CPD techniques, particularly the Pruned Exact Linear Time (PELT) algorithm, in predictive maintenance are often overlooked and need further investigation. PELT, as a state-of-the-art offline CPD method, offers accurate event detection outcomes through its binary segmentation approach and significantly reduces computational complexity compared to exact search methods. The primary advantage of PELT lies in its ability to prune the search space, thereby minimizing computational costs without compromising segmentation accuracy. Given these benefits, this study investigates the effects of PELT's parameters in an industrial context, combining these findings with expert evaluation to validate its applicability in real-world environments [18].

III. Methodology

As discussed, there are several challenges in predictive maintenance, particularly the high costs associated with data collection and data labeling. Given these challenges, integrating advanced CPD techniques into predictive maintenance frameworks presents a promising solution. In this section, CPD methods, with a particular focus on the PELT algorithm with its parameters and the proposed methodology will be discussed.

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The PELT algorithm, introduced by Killick et al., is a powerful CPD method known for its balance between computational efficiency and detection accuracy [14]. PELT is particularly suitable for large-scale industrial applications because it operates in linear time, making it feasible to process extensive datasets in real-time [18].

PELT achieves its efficiency by pruning the search space for potential change points. Traditional change point detection methods often involve exhaustive searches, which can be computationally intensive, especially with large datasets [5]. In contrast, PELT combines dynamic programming with a pruning technique, reducing the number of change points that need evaluation, thereby significantly enhancing scalability and real-time applicability. Alongside its computational benefits, PELT maintains high detection accuracy, effectively identifying multiple change points in large datasets. This is particularly crucial in industrial applications, where early fault detection can prevent downtime and reduce maintenance costs. By detecting change points in vibration data from industrial machinery, PELT enables early identification of equipment failure, allowing maintenance teams to address issues before they escalate, minimizing downtime and maintenance expenses.

A. PELT Models and Key Parameters

PELT operates by minimizing a cost function that includes both the fit of the model to the data and a penalty for the number of change points [11]. The PELT algorithm especially efficient if there is no a-priori data for the number of the change points. This issue is solved via adding a linear penalty function defined as follows,

$$pen(\Gamma) = \beta|\Gamma| \tag{1}$$

where $\beta > 0$ is a smoothing parameter (which we called as penalty value in this work) and Γ corresponds to the best possible segmentation. Here, $|\Gamma|$ is the cardinality of segmentation indices or in other words, the number of the estimated change points. By using the Eq.1, a linear pruning rule can be defined to search each time series sample sequentially for deciding to exclude the potential change point from the set of estimated change points. In this work, we investigate the effects of penalty value, possible cost functions and the minimum segmentation size on the performance.

The ***penalty term*** controls the trade-off between the number of change points detected and the fit of the model. A higher penalty results in fewer change points and vice versa. The optimal penalty needs to be chosen based on the specific application and data characteristics.

The ***cost function*** measures the residual sum of squares, but it can be adapted to other measures such as L_1, L_2 norms, and Radial Basis Function (RBF) depending on the application.

a) L_1 (Linear) : The L_1 model focuses on minimizing the absolute deviations, making it robust to outliers.

b) L_2(Quadratic): The L_2 model minimizes the squared deviations, commonly used for its simplicity and effectiveness in many scenarios.

c) RBF (Radial Basis Function): This model is used for detecting non-linear changes by applying kernel methods to identify changes in the data distribution.

Minimum Segment Size (min_size) parameter defines the minimum length of a segment between two change points. Setting a larger min_size ensures that only significant changes are detected, avoiding false positives from minor fluctuations in the data.

The implementation of PELT in a predictive maintenance framework involves setting these parameters based on the specific characteristics of the machinery and the operational environment.

B. Datasets

Initially, a synthetic data set was created for the study, with predetermined attributes, to facilitate analysis of a known structure. Prior to producing the synthetic dataset, certain parameters are established. The supplied values include the total number of data points (n_samples = 1000) and the number of pre-determined change points (n_change_points = 15). Change points refer to the indices at which the data characteristics change. The values are created as an array of equidistant values. The first element the array is omitted to guarantee that the array starts with the first change point. The starting and ending points of each segment are comprised by randomly generated numbers, which are used to create variability within the data. By specifying a random seed, reproducing these random numbers is guaranteed. A plot of synthetic data is presented in Fig.1. The synthetic dataset consists of 1000 samples with 15 change points. Each segment's start and end points were generated randomly, with added Gaussian noise (mean = 0, standard deviation = 1). The change points were manually annotated and used as ground truth for evaluation.

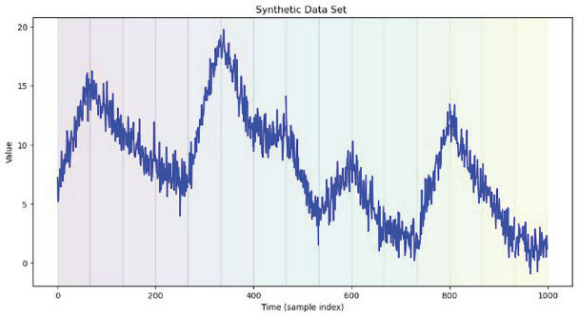

Fig. 1. Synthetic Data Set

To justify the approach used in this study, a real vibration data from industrial environment was gathered from an RTF (Radiant Tube Furnace) in a galvanizing line as depicted in Fig. 2. The combustion air fan of an RTF in a galvanizing line plays a crucial role. This fan supplies the furnace with fresh air necessary for the combustion process. Oxygen is required for the fuel inside the furnace to burn, and the combustion air fan delivers this oxygen to the furnace.

Fig. 2. RTF Fan

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This ensures that the combustion process occurs efficiently and that the furnace reaches the desired temperature. The power of the motor is 45 kW with nominal speed of 2942 rpm. The motor is driven by frequency converter. There are four 12.5kHz vibration sensor located on the asset. The data was collected during the escalation of a bearing fault on the fan side. Subsequently, the bearing was replaced, and a decrease in vibration levels was observed. The collected data is shown in Fig. 3.

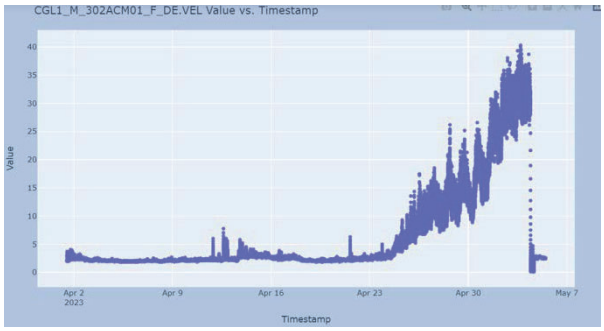

Fig. 3. RTF Fan Vibration Trends

IV. Numerical Results

We evaluated the performance of three different cost models (L_1, L_2, and RBF) using various parameter combinations, including minimum segment size (min_size), penalty value (pen), jump parameter (jump), and tolerance value (tol) as shown in the Fig.4.

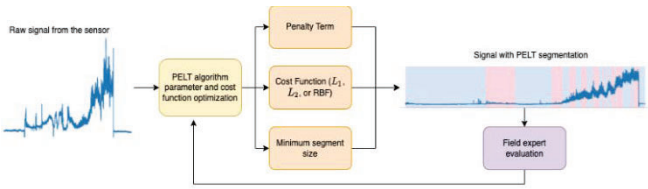

Fig. 4. Workflow of the Appoach for Change Point Detection in Sensor Data

It is evident that algorithm results cannot be accurately evaluated at precisely calculated change points, and therefore, the most optimal tolerance value should be used to create suitable windows. Recall and precision are considered essential evaluation metrics due to their roles in assessing the algorithm's performance. Recall measures the algorithm's ability to correctly identify true positives (actual faults), ensuring that most actual change points are detected. Precision, on the other hand, evaluates the proportion of true positives among all detected change points, indicating the algorithm's effectiveness in minimizing false positives. Balancing these metrics is crucial for developing a robust CPD algorithm that reliably detects faults while avoiding unnecessary false alarms, ultimately enhancing the efficiency and reliability of predictive maintenance systems.

The mathematical formulations for recall and precision defined as follows:

$$Recall = \frac{(TP)}{(FN) + (TP)} \tag{2}$$

$$Precision = \frac{(TP)}{(TP) + (FP)} \tag{3}$$

where TP, FN, and FP correspond true positive, false negative and false positive respectively.

Faults typically develop gradually so fewer change points are preferable. Initially, the correlation between tolerance values and precision was examined, with a tolerance of 20 selected for its higher precision and recall compared to 1, 5, and 10, as shown in Fig. 5. Although the jump value had minimal impact on precision (Fig. 6), a detailed analysis confirmed the robustness of using a jump value of 20. This value improves computational efficiency and detection accuracy. For min_size, a value of 30 was found to provide the highest precision, making it optimal for further filtering (Fig. 7). Finally, the penalty (pen) value's impact on precision, recall, and detected change points was empirically evaluated, considering the unknown true change points.

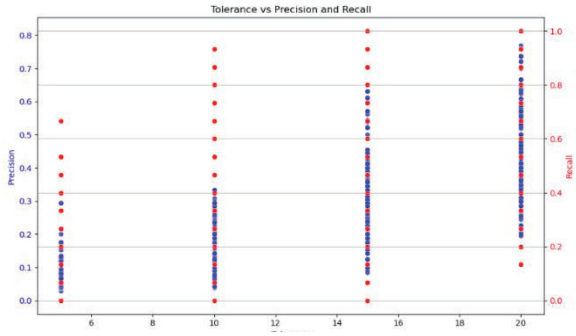

Fig. 5. Plot of Tolerance vs Precision and Recall

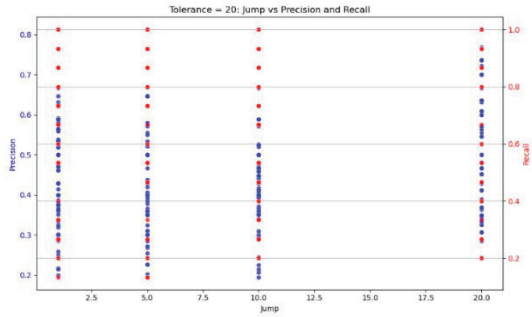

Fig. 6. Plot of Jump vs Precision and Recall with tol=20

We systematically varied the pen values to observe their impact on the change point detection algorithm's performance metrics. As shown in Fig. 8, different penalty settings influence precision, recall, and the total count of detected change points. This evaluation highlights how fine-tuning the penalty parameter can optimize detection, balancing significant change point identification with minimizing false positives. Such optimization is crucial for improving the reliability and accuracy of change point detection in predictive maintenance applications, ensuring robust and effective operational data analysis.

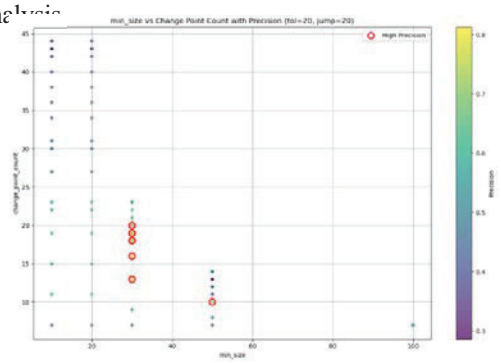

Fig. 7. Relationship Between min_size and Change Point Count with Precision Highlighted

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The results indicate that the L_2 model with a penalty value of 20.0 achieves high precision (0.74) and the highest recall (0.93), suggesting its effectiveness in detecting most true change points while maintaining a reasonable level of precision. Selecting a penalty value of 20.0 offers several advantages. Firstly, it ensures the detection of significant change points without being overwhelmed by minor fluctuations, as evidenced by the model detecting a moderate number of change points which is 19.

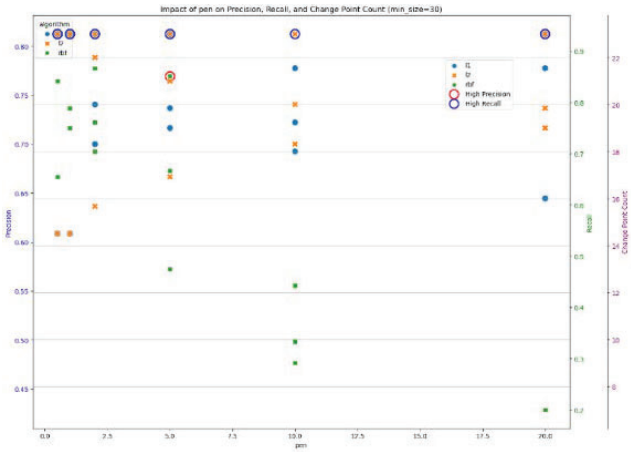

Fig. 8. Analyzing the Influence of Penalty Parameter on Precision, Recall, and Change Point Detection

This focus on critical changes is particularly relevant for maintenance purposes as discussed. Additionally, a higher penalty value helps minimize false positives by reducing the number of detected change points, thus avoiding overfitting to noise and minor variations in the data for further applications.

We also analyzed the actual vibration data obtained from the RTF in a galvanizing line. To evaluate the effectiveness of the PELT algorithm in detecting change points in this real-world dataset, we utilized the min_size and jump values derived from our synthetic data experiments, which are 30 and 20, respectively. We varied the penalty (pen) value from 200 to 700 to analyze its impact on CPD. An essential heuristic consideration is that with larger datasets, a higher penalty value may be required to avoid overfitting and ensure that only significant change points are detected. This is because larger datasets typically have more noise, and a higher penalty helps to filter out minor fluctuations, focusing on substantial changes [20]. To visualize the impact of different penalty values on change point detection, we generated graphs for each pen value as shown in Fig. 9. Lower penalty values result in higher sensitivity, detecting more change points but potentially increasing false positives. Higher penalty values reduce the number of detected change points, focusing on the most significant changes and improving precision as shown in Fig. 10. For a penalty value of 700, the PELT algorithm demonstrated relatively better performance validated by the maintenance experts. Maintenance experts from the field noted that the significant change points identified matched the historical patterns of fault development, providing practical validation for the algorithm's effectiveness. In conclusion, the PELT algorithm, with appropriately tuned parameters, provides a robust and efficient solution for CPD in vibration data. This approach ensures the reliability and operational efficiency of industrial systems, providing valuable insights for maintenance planning and execution.

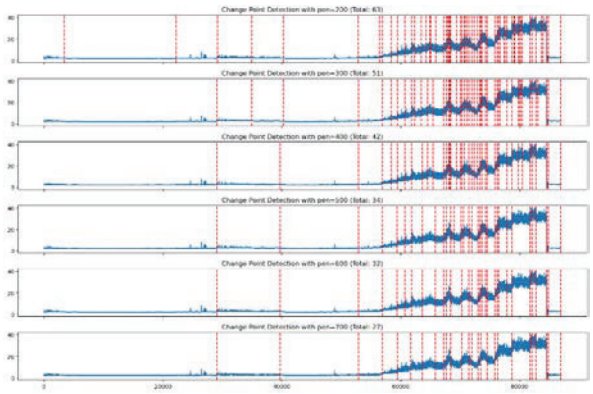

Fig. 9. Impact of different penalty values on change point detection

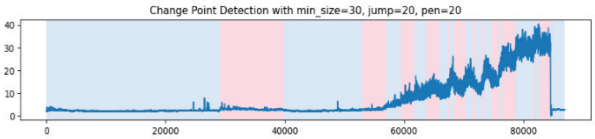

Fig. 10. Change Point Detection with pen=700

V. Conclusion

In this study, we demonstrated the effectiveness of the PELT algorithm in detecting change points in both synthetic and actual vibration data. By optimizing the relevant parameters, we achieved a balance in state detection performance on a realistic data confirmed by the field experts. The parameters were fine-tuned to ensure the algorithm accurately detected faults while minimizing false alarms. Our findings show the potential of advanced CPD methodologies (such as PELT) to significantly improve predictive maintenance operations, offering valuable insights for deployment in real-world industrial scenarios. Future work should focus on refining the PELT parameters for different industrial applications and integrating this approach with other advanced monitoring techniques to further enhance predictive maintenance outcomes. By doing so, we will facilitate the automatic segmentation of massive unlabeled datasets, making the data more useful and significantly improving the efficiency and effectiveness of predictive maintenance frameworks.

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